

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458862.8712 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.048
Transit depth (Rp/Rs)^2	2.68 +/- 1.58 [%]
Orbital Inclination (inc)	85.07 +/- 2.93
Airmass coefficient 1 (a1)	0.21 +/- 0.14
Airmass coefficient 2 (a2)	1.08 +/- 1.07
Scatter in the residuals of the lightcurve fit is	66.74 %
Best Comparison Star	#2 - [260, 93]
Optimal Aperture	1.82
Optimal Annulus	3.0
Transit Duration (day)	0.083 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.048 vs 0.1368 (deviation 0.46σ)
Duration fit vs published	0.083 d vs 0.0758 d (deviation 0.34σ)
Mid-time O–C	-5.5 min
Depth SNR	2.8
Residual scatter	66.74 %

Light curve

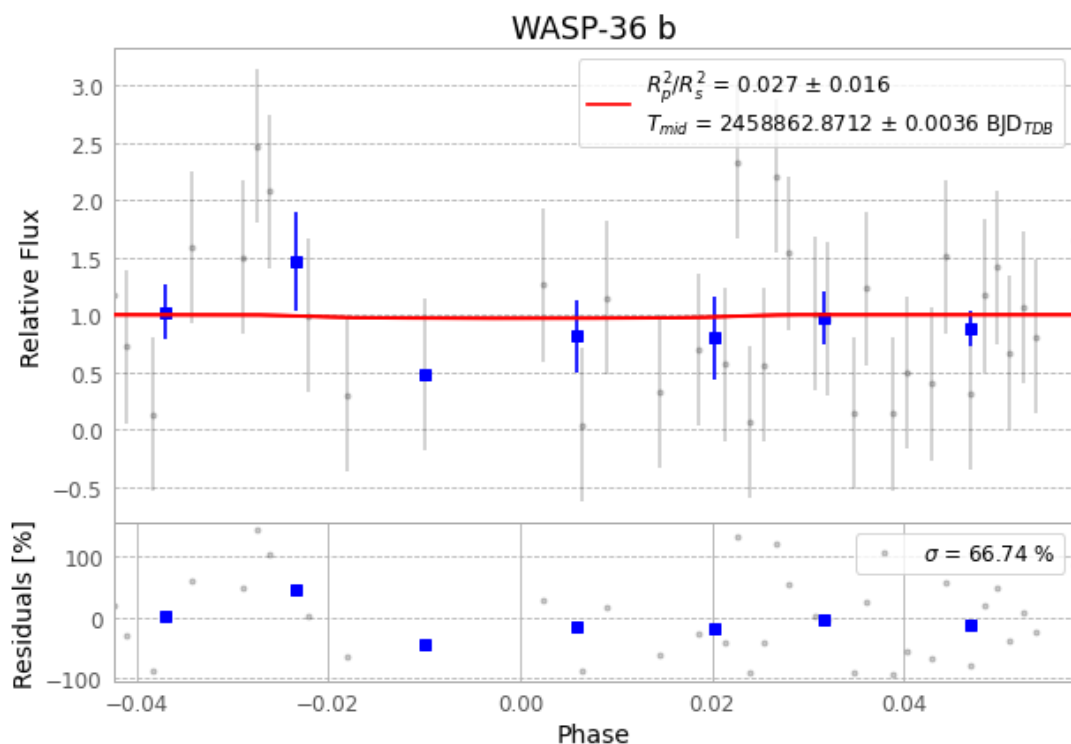


Figure 1 — FinalLightCurve_WASP-36 b_2020-01-14.png (SHA-256 ef2af90756d094e7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [209, 443], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4ec8de6185649983...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-36200114070612.FITS → WASP-36200114105412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-200114.log (b28916237246...), FinalLightCurve_WASP-36 b_2020-01-14.png (ef2af90756d0...), FinalLightCurve_WASP-36 b_2020-01-14.csv (f8716ce4f42b...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 00:49Z.